
Introduction

Discrete Mathematics and Data

Data Science is a relatively new subject. For the most part it is about applications that involve data. Some datasets are small, like the data for the people in this room. Some datasets are large, like for the people who use the internet. Some of the goals are to describe data or to understand data. Sometimes this will involve making a picture and other times it will involve some discrete mathematics.

Discrete Mathematics is everything except for calculus. We will learn about logic and recursion, sets and functions.

In the second portion of the course we will learn about linear algebra. The simplest structure for data is a long list, which we will call a vector.

Out[]//MatrixForm=

$$\begin{pmatrix} 4 \\ 2 \\ 3 \\ 1 \\ 8 \\ 4 \\ 5 \end{pmatrix}$$

Another structure is a matrix with two indices. One index for the row and one for the column.

Out[]//MatrixForm=

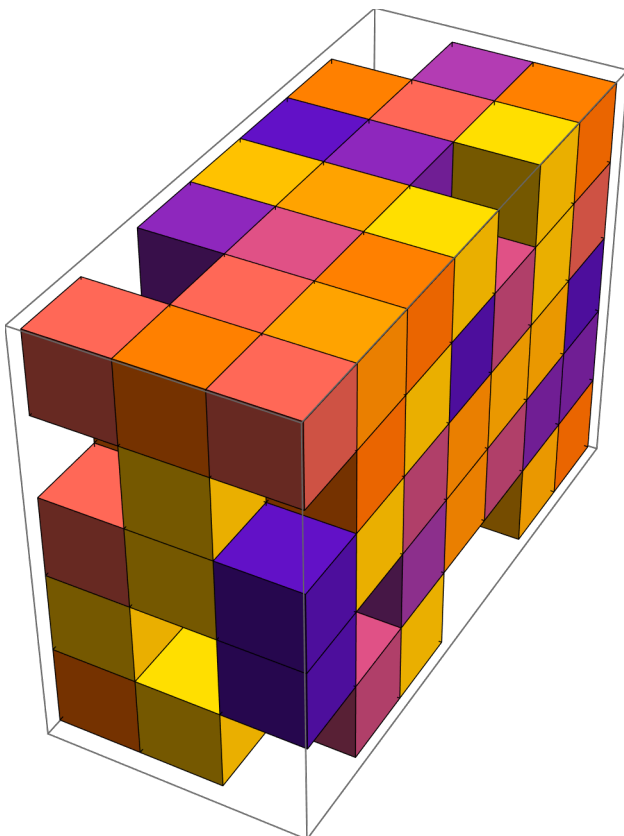
$$\begin{pmatrix} 3 & 5 & 3 & 3 & 3 & 9 & 5 \\ 1 & 2 & 1 & 1 & 6 & 1 & 4 \\ 9 & 2 & 6 & 7 & 6 & 0 & 1 \\ 0 & 5 & 6 & 1 & 5 & 4 & 5 \\ 1 & 8 & 5 & 7 & 0 & 8 & 6 \end{pmatrix}$$

When there are more indices an array they are harder to visualize.

Out[]//MatrixForm=

$$\begin{pmatrix} \begin{pmatrix} 0 \\ 3 \\ 6 \end{pmatrix} & \begin{pmatrix} 6 \\ 5 \\ 9 \end{pmatrix} & \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} & \begin{pmatrix} 8 \\ 7 \\ 9 \end{pmatrix} & \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix} & \begin{pmatrix} 0 \\ 5 \\ 7 \end{pmatrix} & \begin{pmatrix} 5 \\ 6 \\ 5 \end{pmatrix} \\ \begin{pmatrix} 4 \\ 2 \\ 5 \end{pmatrix} & \begin{pmatrix} 3 \\ 9 \\ 9 \end{pmatrix} & \begin{pmatrix} 6 \\ 6 \\ 4 \end{pmatrix} & \begin{pmatrix} 4 \\ 8 \\ 1 \end{pmatrix} & \begin{pmatrix} 0 \\ 8 \\ 9 \end{pmatrix} & \begin{pmatrix} 6 \\ 8 \\ 6 \end{pmatrix} & \begin{pmatrix} 0 \\ 9 \\ 0 \end{pmatrix} \\ \begin{pmatrix} 1 \\ 8 \\ 1 \end{pmatrix} & \begin{pmatrix} 3 \\ 1 \\ 8 \end{pmatrix} & \begin{pmatrix} 8 \\ 0 \\ 8 \end{pmatrix} & \begin{pmatrix} 2 \\ 1 \\ 7 \end{pmatrix} & \begin{pmatrix} 4 \\ 4 \\ 4 \end{pmatrix} & \begin{pmatrix} 8 \\ 8 \\ 9 \end{pmatrix} & \begin{pmatrix} 5 \\ 9 \\ 1 \end{pmatrix} \\ \begin{pmatrix} 8 \\ 6 \\ 2 \end{pmatrix} & \begin{pmatrix} 2 \\ 8 \\ 2 \end{pmatrix} & \begin{pmatrix} 0 \\ 0 \\ 4 \end{pmatrix} & \begin{pmatrix} 0 \\ 3 \\ 7 \end{pmatrix} & \begin{pmatrix} 7 \\ 8 \\ 3 \end{pmatrix} & \begin{pmatrix} 8 \\ 3 \\ 0 \end{pmatrix} & \begin{pmatrix} 9 \\ 0 \\ 1 \end{pmatrix} \\ \begin{pmatrix} 2 \\ 2 \\ 6 \end{pmatrix} & \begin{pmatrix} 2 \\ 4 \\ 8 \end{pmatrix} & \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} & \begin{pmatrix} 3 \\ 7 \\ 0 \end{pmatrix} & \begin{pmatrix} 9 \\ 0 \\ 9 \end{pmatrix} & \begin{pmatrix} 7 \\ 2 \\ 4 \end{pmatrix} & \begin{pmatrix} 6 \\ 9 \\ 0 \end{pmatrix} \end{pmatrix}$$

Out[]:=



Often it will make sense to consider systems of equations from the entries of a matrix, or weighted sums of entries from a tensor, both of which are covered in Linear Algebra.

One of the basic tools, both in linear algebra and in discrete mathematics is summation.

Logic and Proofs

All men are mortal.

Socrates is a man.

Therefore Socrates is mortal.

Translating from words to symbols, we can consider this in propositional logic

$A \Rightarrow B$

$C \Rightarrow A$

so we conclude that $C \Rightarrow B$

or in logic with quantifiers. Here $\forall$ means “For All”, and $\in$ means “element of”

$\forall x \in S, P(x)$ where $P(x)$ is the property that “x is mortal”

$\text{Socrates} \in S$

so we conclude $P(\text{Socrates})$

Knights always tell the truth. Knaves always lie.

There are two people A and B. Can you tell if they are knights or knaves?

A says “Both of us are knights.” and B says, “A is a knave”.

In Calculus, we have seen ϵ - δ definitions of the limit.

For all ϵ , there is a δ , such that $P(\epsilon, \delta)$

There are theorems like the Fundamental Theorem of Calculus, which have hypotheses and conclusions.

Theorems require proofs. There are different levels of rigor. We want an airtight fact, that even the most stubborn skeptic will have to agree is correct.

Sums and Numbers

We will investigate numbers and their properties.

12 divides 108

17 is a prime number

In discrete math, the analogy for the integral is the sum, usually denoted with the Σ symbol.

How do we do this one?

$$\text{Out}[n]=\sum_{k=0}^n k$$

In data science, sums are pretty common. Also in linear algebra.

How many zeros at the end of 100!

Sets and Functions

We are used to functions from calculus, e.g. $f(x) = \sin(x)$, where we plug in a number for x and get another number. For every x, there is one y.

In a dataset, a function could be between any two sets. They don't have to be numbers.

Imagine a dataset including people and cars. Further imagine that each car belongs to a person. Then there is a function $f(\text{car}) = \text{owner of that car}$.

It might be that a person owned more than one car, or that some people didn't own a car. Then there wouldn't be an inverse function from people to cars. But if every person owned exactly one car then there would be an inverse function, and the number of cars would have to be the same as the number of people.

If $A = \text{freshmen}$ and $B = \text{people from Maryland}$ then what is

$A \cup B$

$A \cap B$

$A - B$

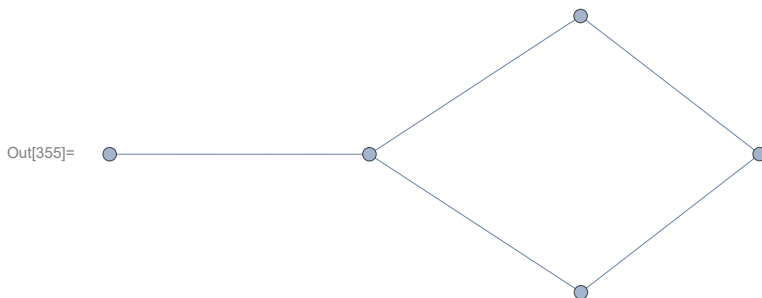
What is the Venn diagram?

Graphs and Discrete Math

We are used to some types of maps where locations are connected by lines. These are examples of graphs. Or the nerves connecting parts of the body to the brain.

We won't go too far with graphs, but they are a good example of discrete structures and have lots of applications in data science. Some pieces of data even are graphs, like road maps.

In[355]:= **RandomGraph**[{5, 5}]



Vectors, Matrices, Tensors

Applications of Linear Algebra

Optimization with least squares

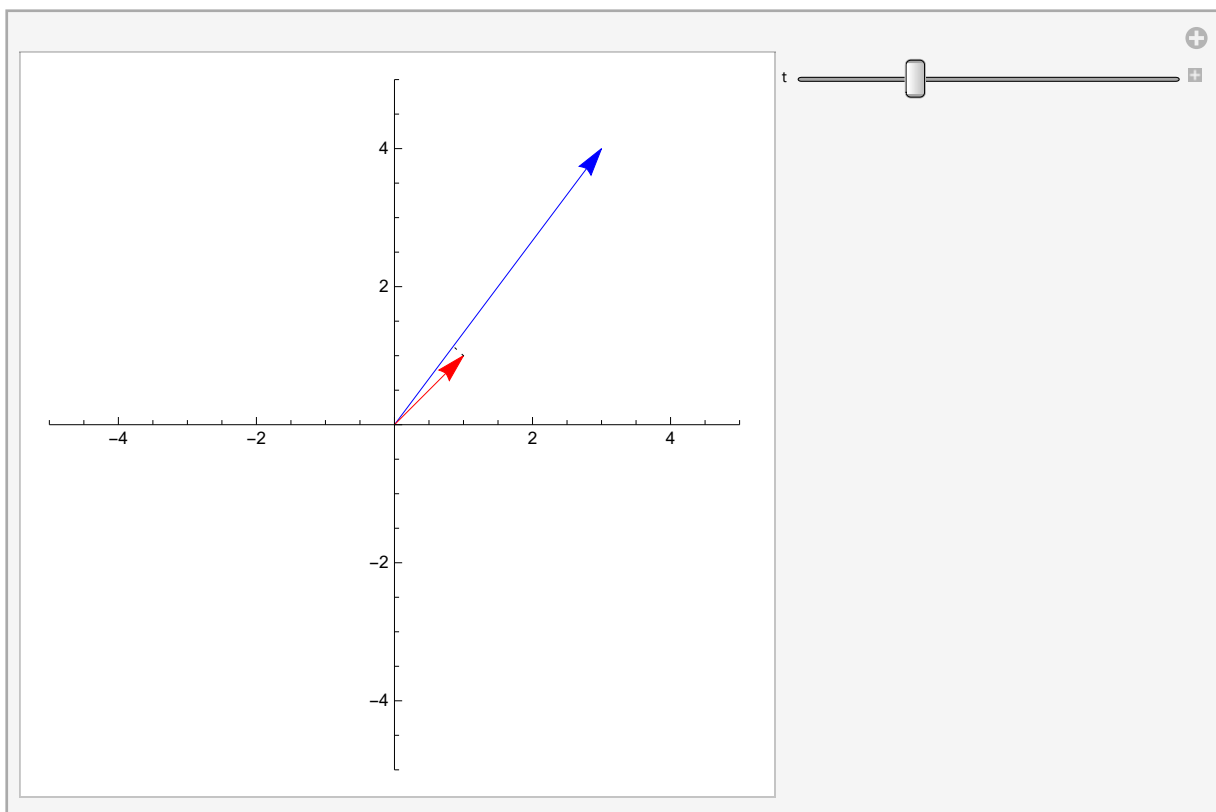
Minimize the error:

$$x = 3t = 1$$

$$y = 4t = 1$$

the error-squared is $(x - 1)^2 + (y - 1)^2$

Out[]=



Minimize the error:

$$\alpha_1 t = 0$$

$$\alpha_2 t = 0$$

$$\alpha_3 t = 0$$

Out[]=

